

2. Technical Knowledge

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Attempted: 0/30

Question 1

Revisit Later

A full-duplex channel is having a bandwidth of 10 kHz. How much maximum bits can we send at a time in one way communication?

Select an option

- ☐ 10,000 bits
- ☐ 20,000 bits
- ☐ 40,000 bits
- ☐ 50,000 bits

Question 1[Revisit Later](#)

A full-duplex channel is having a bandwidth of 10 kHz. How much maximum bits can we send at a time in one way communication?

Select an option☐ 10,000 bits☐ 20,000 bits☐ 40,000 bits☐ 50,000 bits

2. Technical Knowledge

< 1 2 3 4 5 6 7 8 9 10

Attempted: 0/30

Question 2

Revisit Later

A person is trying to send some coded messages by rolling two dices. Every time a message is sent after getting a sum as 9. The probability of getting a sum as 9 is $1/9$, what will be the information content of these messages?

Select an option

☐ 3 bits☐ 4 bits☐ 6 bits☐ 9 bits

Question 3

Revisit Later

Find the relative permeability of the air when the reluctance of a rectangular magnetic circuit is 10 At/Wb and the area of the circuit is 200 m^2 .

Select an option

☐ 0.1☐ 1☐ 0.001☐ 0.001

Attempted: 0/30

Question 4

Revisit Later

Match the conditions which are true for the formation of a virtual channel in NMOS?

A. Condition for saturation current in NMOS transistor	1. $V_{gs} < V_{tn}$
B. Condition for linear current in NMOS transistor	2. $V_{ds} < (V_{gs} - V_{tn})$
C. Condition when no virtual channel created	3. $V_{ds} \geq (V_{gs} - V_{tn})$

Select an option

☐ C - 1, B - 2, A - 3☐ C - 4, B - 3, A - 2☐ C - 1, B - 3, A - 2☐ C - 2, B - 1, A - 3

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Question 5

Revisit Later

Consider the given specifications for a bipolar junction transistor in common emitter configuration.

$$\alpha = 0.995$$

Collector current = 4 A

Find the emitter current.

Select an option

☐ 4.99 A☐ 4.02 A☐ 3.98 A☐ 248.75 mA

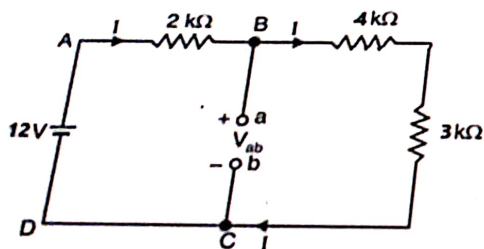
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Question 6

Revisit Later

Select an option

Find the voltage across V_{ab} in the given circuit using Kirchhoff's voltage law.

☐ 8.54 V☐ 9.34 V☐ 8.34 V☐ 7.32 V

Question 7

Revisit Later

Which input is provided for the frequency response analysis of the system?

Select an option

☐ Unit step

☐ Ramp

☐ Sinusoidal

☐ Delta function

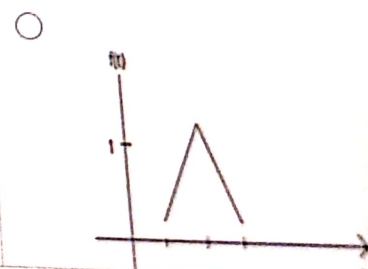
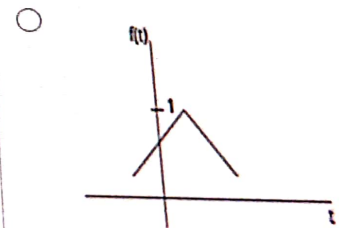
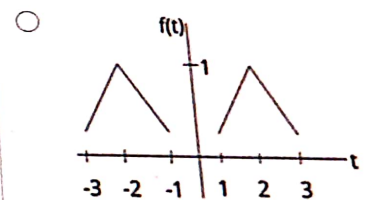
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Question 8

🔖 Revisit Later

Which of the given options represents a causal system?

Select an option



Question 9

Revisit Later

Fill in the blank.

The "rise time" of a pulse denotes the time frame to go from _____.

Select an option

- ☐ the base line to the maximum high voltage
- ☐ 10% of the pulse amplitude to the maximum
- ☐ the base line to 90% of the pulse amplitude
- ☐ 10% of the pulse amplitude to 90%

2. Technical Knowledge

Question 9

Fill in the blank.

The "rise time" of a pulse denotes the time frame to go from _____.

Revisit Later

Attempted: 0/30

Select an option

- ☐ the base line to the maximum high voltage
- ☐ 10% of the pulse amplitude to the maximum high voltage
- ☐ the base line to 90% of the pulse amplitude
- ☐ 10% of the pulse amplitude to 90% of the pulse amplitude

2. Technical Knowledge

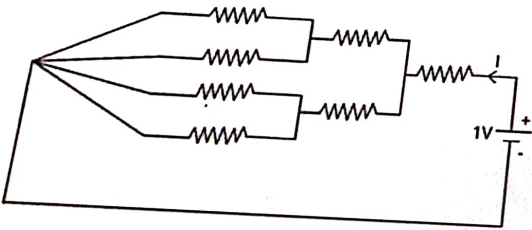
5 6 7 8 9 10 11 12 13 14

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Question 10

Revisit Later

In the figure given below, all the resistances have a value of $2\ \Omega$ each. Find the value of I in this scenario.



Select an option

☐ 1/7 A☐ 2/7 A☐ 4/7 A☐ 5/7 A

2. Technical Knowledge

< 6 7 8 9 10 11 12 13 14 15 >

Attempted: 0/30

Question 11

Revisit Later

For a ground wave having a wavelength of 200 m and the height of the transmitting antenna as 100 m, the radius of the earth is given as 6400 Km. What will be the transmitting range of an antenna?

Select an option

☐ 128×10^4 Km☐ 129×10^5 Km☐ 127×10^4 Km☐ 129×10^4 Km

Question 12

Revisit Later

A sinusoidal carrier wave is amplitude modulated by another sinusoidal wave to a depth of 20%. A third sinusoidal modulating signal modulates it to a depth of 30% and a fourth one modulates it to a depth of 40%. Calculate the final percentage of modulation.

Select an option

☐ 30.13%☐ 46.64%☐ 65.43%☐ 53.85%☐ 90%

Question 13

Revisit Later

Find the unity power factor frequency of an RLC circuit having an inductor value of 2H and a capacitor value of 20 microfarads .

Select an option

☐ 25Hz☐ 24Hz☐ 50Hz☐ 20Hz

2. Technical Knowledge

Question 14

For the given NMOS differential pair with a common-mode voltage (V_{cm}) applied, find the overdrive voltage (V_{ov}) and the voltage between the gate and source terminal (V_{gs}). Assume the given parameters for the circuit.

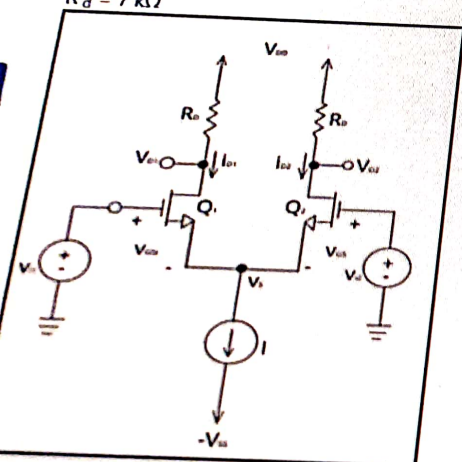
$$V_{dd} = 5V$$

$$K_n'(W/L) = 2 \text{ (mA/V}^2\text{)}$$

$$V_t = 0.99V$$

$$I = 0.62mA$$

$$R_d = 7 \text{ k}\Omega$$



Revisit Later

Select an option

☐ $V_{ov} = 0.556 \text{ V}, V_{gs} = 1.546 \text{ V}$

☐ $V_{ov} = 0.123 \text{ V}, V_{gs} = 0.345 \text{ V}$

☐ $V_{ov} = 0.988 \text{ V}, V_{gs} = 3.233 \text{ V}$

☐ $V_{ov} = 2.234 \text{ V}, V_{gs} = 0.543 \text{ V}$

< 10 11 12 13 14 15 16 17 18 19 >

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Question 15

Revisit Later

For the T.F. $G(s) = 144/4s^2 + 12s + 144$, find the natural frequency and damping ratio respectively.

Select an option

☐ 6, 0.5☐ 12, 0.5☐ 12, 1☐ 6, 0.25

Question 16

 Revisit Later

Find the steady-state error for the signal $[5 + 2t + 7t^2] u(t)$, which passes through a unit feedback system.

Select an option

☐ 0☐ 7☐ 14☐ 2

Revisit Later

Select an option

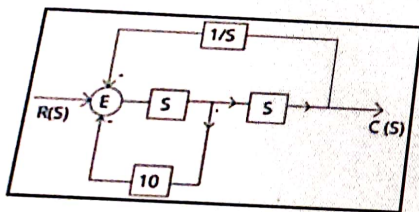
☐ $s^2/(1+10s)$

☐ $s/(1+10s)$

☐ $s/(1+11s)$

☐ $s^2/(1+11s)$

Question 17

Find the transfer function $T(s)$ for the following system.

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Test Time: 0

2. Technical Knowledge

13 14 15 16 17 18 19 20 21 22

Attempted: 0/30

Question 18

Revisit Later

An SCR (Silicon Controlled Rectifier) has a one cycle surge current of 4000 A for a 50 Hz supply. What is its half-cycle surge supply?

Select an option

- ☐ 2000 A
- ☐ 8000 A
- ☐ 2828.43 A
- ☐ 5656.85 A

Question 19

Revisit Later

What does a high torque to weight ratio indicate in an analog indicating instrument?

Select an option

- ☐ Low frictional loss
- ☐ No frictional loss
- ☐ High frictional loss
- ☐ Moderate frictional loss

2. Technical Knowledge

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Test Time: 0

< 15 16 17 18 19 20 21 22 23 24

Attempted: 0/30

Question 20

What is the value of real power through an inductor?

Revisit Later

Select an option

☐ 0

☐ VI

☐ 1

☐ -1

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< 16 17 18 19 20 21 22 23 24 25 >

Attempted: 0/30

Question 21

Revisit Later

Calculate the time period of the A.C. circuit if it is operating at a frequency of 50 Hz.

Select an option

☐ 20 ms

☐ 50 ms

☐ 2 s

☐ 5 s

Question 22

Revisit Later

There is a complicated network with 23 branches that consists of 12 nodes. Based on this statement, how many equations shall be required for the node-voltage method?

Select an option

☐ 23☐ 11☐ 12☐ 35

Question 23

[Revisit Later](#)

In a two-port network, what will be the image impedance if the open circuit and short circuit impedances are 180 ohms and 20 ohms, respectively?

Select an option

☐ 180 ohms☐ 60 ohms☐ 90 ohms☐ 200 ohms

Question 24

[Revisit Later](#)

The open-loop transfer function of a system is $3.25(s+2) / (s^3 + Ps^2 + 3s + 2)$. If the system has sustained oscillations of 2.5 rad/s, determine the value of P. Assume there are no roots in RHP.

Select an option

☐ P = 1.36☐ P = 1.86☐ P = 0.77☐ P = 0.97

2. Technical Knowledge

< 20 21 22 23 24 25 26 27 28 29

Attempted: 0/30

Question 25

Revisit Later

A 3 phase, 220 V, 50 Hz induction motor of 15 kW power with 6 poles, drives a constant torque load and has a speed of 840 RPM. Calculate the new operating speed if there is a 20% drop in voltage and a 10% drop in frequency. (Assume that all losses remain constant.)

Select an option

☐ 773 RPM☐ 787 RPM☐ 754 RPM☐ 768 RPM

Question 26

 Revisit Later

A field coil has a heat dissipating surface of 0.5 m^2 and the length of the mean turn as 1 m . It dissipates a loss of 170 W , the emissivity being 30 W/m^2 . Calculate the final steady temperature rise of the coil.

Select an option

☐ 30°C ☐ 10°C ☐ 40°C ☐ 20°C

Question 27

[Revisit Later](#)

In a power generation setup, we have a Pelton wheel turbine. It is connected to an alternator operating at a frequency of 60 Hz having 12 poles. Calculate the speed of the Pelton wheel.

Select an option

☐ 450 rpm☐ 300 rpm☐ 600 rpm☐ 800 rpm

2. Technical Knowledge

21 22 23 24 25 26 27 28 29 30

Attempted: 0/30

Question 28

Revisit Later

Which of the following statements is true with reference to the superposition theorem?

Select an option

- ☐ For all the present sources, every source is considered one at a time and replaced by their internal impedance.
- ☐ All sources are considered independently.
- ☐ All sources are considered together simultaneously.
- ☐ For all the present sources, every source is considered one at a time and replaced by their internal resistance.

th reference to the superposition theorem?

Select an option

- ☐ For all the present sources, every source is considered one at a time with all others replaced by their internal impedance.
- ☐ All sources are considered independently.
- ☐ All sources are considered together simultaneously.
- ☐ For all the present sources, every source is considered one at a time with all others replaced by their internal resistance.

2. Technical Knowledge

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Test

21 22 23 24 25 26 27 28 29 30

Attempted: 0/30

Question 29

Revisit Later

While selecting the insulator for transmission cables, an experiment was conducted to calculate its dielectric constant. It was placed in between a parallel plate capacitor with a capacitance of $2.00 \mu\text{F}$, completely occupying the space between the two plates having the area of the plates to be 1.25 m^2 . The distance between the plates is 0.005 m . What is the dielectric constant of the insulator?
Given - $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

Select an option

☐ 812 F/m

☐ 904 F/m

☐ 659 F/m

☐ 869 F/m

2. Technical Knowledge

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Test

21 22 23 24 25 26 27 28 29 30

Attempted: 0/30

Question 30

Revisit Later

The transfer function of a plant with a unity negative feedback control system is $50/(s^2 - 20)$. Find the natural frequency of the system for the following specifications.
Settling Time (for 5% tolerance band) = 75 ms
Peak Overshoot = 10%

Select an option

☐ 33.34 kHz

☐ 67.67 kHz

☐ 67.67 Hz

☐ 33.34 Hz

